

Analysis 3: Within-Patient Correlation (Repeated Measures)

GEE and Penalized Time-Varying Cox Models on the Hourly Panel

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0.1 Overview

The longitudinal report (step 09) never modelled the within-patient correlation of repeated hourly measurements. This script addresses that gap in two classical ways: (1) GEE with an exchangeable working correlation, and (2) a penalized (ridge) time-varying-covariate Cox model on a counting-process formulation. Both operate on the sampled long panel (2,500 stays) for tractability.

0.2 Load Libraries and Set Seed

```
suppressPackageStartupMessages({ library(geepack); library(sandwich); library(survival) })
set.seed(486649)
```

0.3 Locate and Read Data

```
find_data <- function(f) {
  cands <- c(file.path("../..//processed_data", f),
             file.path("processed_data", f), f)
  hit <- cands[file.exists(cands)][1]
  if (is.na(hit)) stop("cannot find ", f); hit
}
OUT <- dirname(find_data("08_panel_sample_long.parquet"))
d0 <- as.data.frame(arrow::read_parquet(find_data("08_panel_sample_long.parquet"))) # full copy
d <- d0
```

0.4 Define the Dynamic Incident Outcome

The stored per-hour label is nearly constant within a patient, so a random intercept would suffer complete separation. A proper early-warning target restricts to at-risk hours and defines the outcome as whether sepsis onset occurs within the next 12 hours. This outcome varies within septic patients, making it a well-posed repeated-measures formulation.

```
HORIZON <- 12
d <- d[is.na(d$onset_hour) | d$hour < d$onset_hour, ]
d$y <- as.integer(!is.na(d$onset_hour) &
                  (d$onset_hour - d$hour) <= HORIZON &
                  (d$onset_hour - d$hour) > 0)
cat(sprintf("Loaded sampled panel (at-risk hours): %d stay-hours, %d stays, incident-within-12h
            nrow(d), length(unique(d$stay_id)), mean(d$y)))
```

```
## Loaded sampled panel (at-risk hours): 76538 stay-hours, 2254 stays, incident-within-12h rate
```

0.5 Standardize Predictors

Median imputation and z-scoring yield odds ratios on a comparable scale across predictors.

```
zt <- function(x) { x[is.na(x)] <- median(x, na.rm=TRUE); as.numeric(scale(x)) }
d$z_hr <- zt(d$hr); d$z_map <- zt(d$map); d$z_rr <- zt(d$resp_rate)
d$z_temp <- zt(d$temp_c); d$z_lac <- zt(d$lactate); d$z_hour <- zt(d$hour)
preds <- c("z_hr", "z_map", "z_rr", "z_temp", "z_lac", "z_hour")
form <- as.formula(paste("y ~", paste(preds, collapse=" + ")))
```

0.6 GEE vs Naive Pooled GLM

A per-patient random-intercept GLMM is degenerate here: most patients never experience an event, so the per-cluster intercepts diverge and the variance component explodes. GEE avoids this by estimating population-averaged effects with an exchangeable within-patient working correlation and cluster-robust sandwich standard errors. Contrasting with a naive pooled GLM that treats the approximately 76,000 stay-hours as independent quantifies how much ignoring within-patient correlation understates the standard errors.

```
d <- d[order(d$stay_id, d$hour), ]
g_pool <- glm(form, data=d, family=binomial)

se_naive <- sqrt(diag(vcov(g_pool)))
se_robust <- sqrt(diag(vcovCL(g_pool, cluster = d$stay_id, type = "HCO")))
cat("fitting GEE (exchangeable within-patient correlation) ...\n")

## fitting GEE (exchangeable within-patient correlation) ...

g_gee <- geeglm(form, id=stay_id, data=d, family=binomial, corstr="exchangeable")
alpha <- summary(g_gee)$corr[["Estimate"]]
cat(sprintf("exchangeable within-patient correlation alpha = %.3f\n", alpha))

## exchangeable within-patient correlation alpha = 0.676

nm <- names(coef(g_pool))
comp <- data.frame(
```

```

term          = nm,
OR_glm        = round(exp(coef(g_pool)), 3),
SE_naive      = round(se_naive[nm], 3),
SE_cluster_robust = round(se_robust[nm], 3))
comp$SE_inflation <- round(comp$SE_cluster_robust / comp$SE_naive, 2)
rownames(comp) <- NULL
cat(sprintf("\nCluster-robust vs naive SEs on the pooled GLM (within-patient alpha=%.3f):\n", al

##
## Cluster-robust vs naive SEs on the pooled GLM (within-patient alpha=0.676):

print(comp, row.names=FALSE)

##          term OR_glm SE_naive SE_cluster_robust SE_inflation
## (Intercept)  0.014   0.042           0.186           4.43
##          z_hr  1.206   0.021           0.051           2.43
##          z_map  0.880   0.021           0.047           2.24
##          z_rr  1.137   0.019           0.043           2.26
##          z_temp 1.083   0.020           0.046           2.30
##          z_lac  0.845   0.030           0.099           3.30
##          z_hour 0.175   0.043           0.193           4.49

write.csv(rbind(comp,
                 data.frame(term="__alpha_within_patient__", OR_glm=round(alpha,4),
                             SE_naive=NA, SE_cluster_robust=NA, SE_inflation=NA)),
          file.path(OUT, "13_analysis3_gee.csv"), row.names=FALSE)

```

0.7 Penalized Time-Varying-Covariate Cox Model

A counting-process (start, stop] formulation of the hourly data supports a ridge-penalized Cox model that captures how time-varying physiology predicts the hazard of sepsis onset. Cluster-robust variance accounts for within-patient correlation. The full sampled panel (not the pre-onset-only GEE frame) is used here.

```

cat("\nbuilding counting-process time-varying Cox (ridge) ...\n")

##
## building counting-process time-varying Cox (ridge) ...

dc <- d0[order(d0$stay_id, d0$hour), ]
dc$z_hr <- zt(dc$hr); dc$z_map <- zt(dc$map); dc$z_rr <- zt(dc$resp_rate)
dc$z_temp <- zt(dc$temp_c); dc$z_lac <- zt(dc$lactate)
dc$start <- dc$hour
dc$stop <- dc$hour + 1
dc$ev <- as.integer(!is.na(dc$onset_hour) & dc$hour == dc$onset_hour)
dc <- dc[is.na(dc$onset_hour) | dc$hour <= dc$onset_hour, ]
tv <- coxph(Surv(start, stop, ev) ~
            ridge(z_hr, z_map, z_rr, z_temp, z_lac, theta=1) + cluster(stay_id),
            data=dc)
cox_tab <- data.frame(
  term = c("z_hr", "z_map", "z_rr", "z_temp", "z_lac"),

```

```

    HR    = round(exp(coef(tv)), 3))
cat("\nTime-varying Cox (ridge) hazard ratios for incident onset:\n")

##
## Time-varying Cox (ridge) hazard ratios for incident onset:
print(cox_tab, row.names=FALSE)

##      term      HR
##      z_hr 1.301
##      z_map 0.653
##      z_rr 1.064
##      z_temp 1.010
##      z_lac 0.891
write.csv(cox_tab, file.path(OUT, "13_analysis3_tvcox.csv"), row.names=FALSE)
cat("\nAnalysis 3 complete.\n")

##
## Analysis 3 complete.

```